



Calculation Report 0040499642_001 [00]

Subject: VSSP - Tuning Check (TC) – JLR L460, SW: 450_700_C_08_00		
Change Index:		
Handled by:	Bernd Rommel (VM/EYV5-LA)	CAE Task: CAE025726
Client:	Soundariya, B (MS/EYV3-VM)	Part No.: 5432.058.546-01 #431
Date:	06.07.2026	Base report:
Circulation: VM/EYV5-LA, Soundariya, B (MS/EYV3-VM), Koenig Simon (VM/EAE4-CS)		

1 Task

The objective of this analysis was to verify the system stability for the Jaguar Land Rover (JLR) MLA-H MY28 project. A VSSP Tuning Check (TC) was performed using software version 450_700_C_08_00 with hardware library H2E_AEI_A_10_01_C70. Vehicle data for the initial work packages (SWP/1/ and SWP/2/) was recorded using a Range Rover L460 (LHD, iVAR).

2 Results

Requirement VSSP TC	Status
SWP/1/ VT Marginal tune identification completed and marginal dcm provided (no report)	[1]
SWP/2/ VT Plant identification done, and test data provided (no report)	[1]
SWP/3/ CAE Plant identification done and marginal tune VM offsets are in acceptable range	[1]
SWP/4/ CAE Controls identification is done and VM offset validation successful	[1] [2]
SWP/5/ CAE Stability DOE done and BOSCH internal stability requirement VM>0.4 fulfilled	[1] [3]
SWP/6/ VT Critical tune validation done	[1] [4]

[1] no report – [2] VT AUTOSAR tuning cannot be applied to vECU – [3] all stable, some VM<0.4 – [4] stability VM<0.4 not explicitly tested

SWP/1/ & SWP/2/ (Vehicle Testing): Marginal tune and plant identification were successfully completed using an extended test catalogue that included circle driving and hands-on steering conditions.

SWP/3/ (CAE Plant Identification): The marginal tunes identified in vehicle testing result in minimal Vector Margin (VM) offsets, confirming validity of the plant model and allowing to proceed subsequent VSSP steps.

SWP/4/ (CAE Controls Validation): A minor deviation between simulation and vehicle test data was observed in the Nyquist stability analysis. This discrepancy was attributed to a mismatch in AUTOSAR tuning parameters between the virtual ECU (vECU) and the vehicle's marginal tune. As the final software tuning was not modified for subsequent steps, this deviation was not considered a blocker.

SWP/5/ (CAE Stability DOE): The stability DOE confirmed that the system is stable across all tested operating conditions with the 450_700_C_08_00 tuning. However, certain test points violated the internal Bosch requirement of VM > 0.4. These conditions were identified as highly specific and non-representative of real-world driving:

Asphalt: Hands-off, on-center, slow driving (<10 km/h) with a torsion bar torque (TBT) > 1.2 Nm.

Hoist: Hands-off, on-center, slow driving (10-30 km/h) with a TBT > 1.2 Nm.

SWP/6/ (Vehicle Validation): The final on-vehicle validation for the most critical conditions identified in simulation was not performed.

3 Conclusion

The VSSP tuning analysis confirms that software version 450_700_C_08_00 is stable for all operating conditions investigated. While the system violates the internal Bosch stability requirement (VM > 0.4) under a few specific, high-torque, low-speed scenarios, these cases are considered theoretical and do not represent a risk of sustained oscillation in real-world driving. Based on this engineering assessment, the current tuning is judged robust and acceptable for the application.

07.07.2026

 X
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Fulfils requirement / Unclear if requirement fulfilled / Violates requirement



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4 Attachments

4.1 Model-Info

Following model parameters for steering gear 4532.046.644-01 were provided by Soundariya, B (MS/EVS3-VM) and derived from existing toothing data.

Information	
Customer	JLR
Project	MLA High SCU4 D6a
Number of list of parts (steering gear)	4532_046_644_01
Software Label	450_700_C_08_00
Motor	M82S
Manufacturer	Nidec
Dimension	
Max. Rack Travel [mm]	95.7
Rack Diameter [mm]	28
Max. Parking Force [N]	14500
Max. Servo Force [N]	14154
Mass	
Rack [kg]	3.194
KGT Nut [kg]	1.257
Outer Bearing Ring [kg]	0.361
Mass Moment of Inertia	
Motor (shaft with rotor) [kgmm ²]	133.1
Tooth Wheel at Engine Shaft [kgmm ²]	44.76
KGT Package [kgmm ²]	951.94
Steering Pinion [kgmm ²]	19.18
Input Shaft + Sensor [kgmm ²]	14.71
Gear Ratio	
Toothing Ratio [mm/rev]	63.0...72.2
Toothing Number	V17312
Belt [-]	2.4894
KGT pitch [mm/rev]	7
Torsion Bar	
Torsion bar stiffness [Nm/°]	2.5
Max. Torsion bar deflection [deg]	4.8
Design KGT	
KGT Thread Generation	3
Outer BNG spring	WF_243
Installation space outer BNG spring [mm]	0.2
Inner BNG spring	WF_260
Installation space inner wave spring [mm]	0.023
Bearing Type	Double Row / Pressure Angle 40[deg]
Friction	
Friction Motor [Ncm]	2.755
Friction NUT [Ncm]	12
Yoke spring force [N]	260
Yoke Contact angle [°]	45



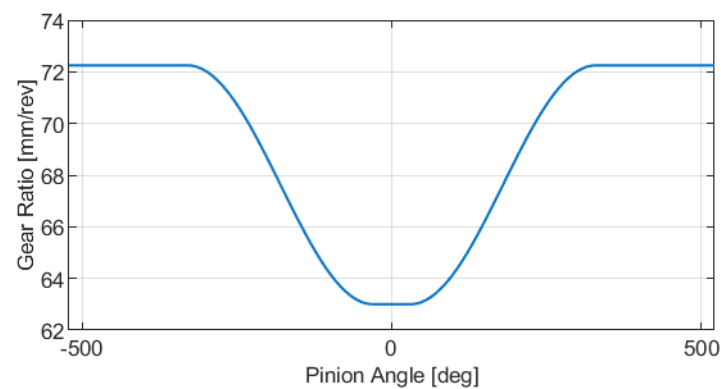
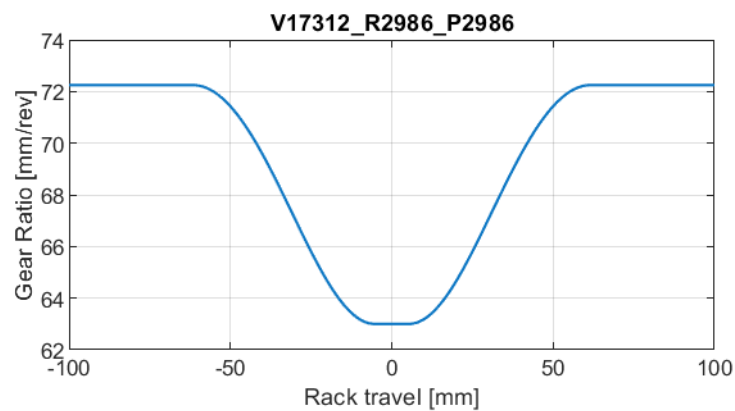
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4.1.1.1 Rack-Pin-Gear Transmission

The following rack-pin-gear toothing data was extracted from [technical report RS146893](#).

ITE calculation name:

Sensor Toothing	Values
rack number	R2986
pinion number	P2986
VERA calculation	V17312
ratio [mm/rev]	63.0...72.2

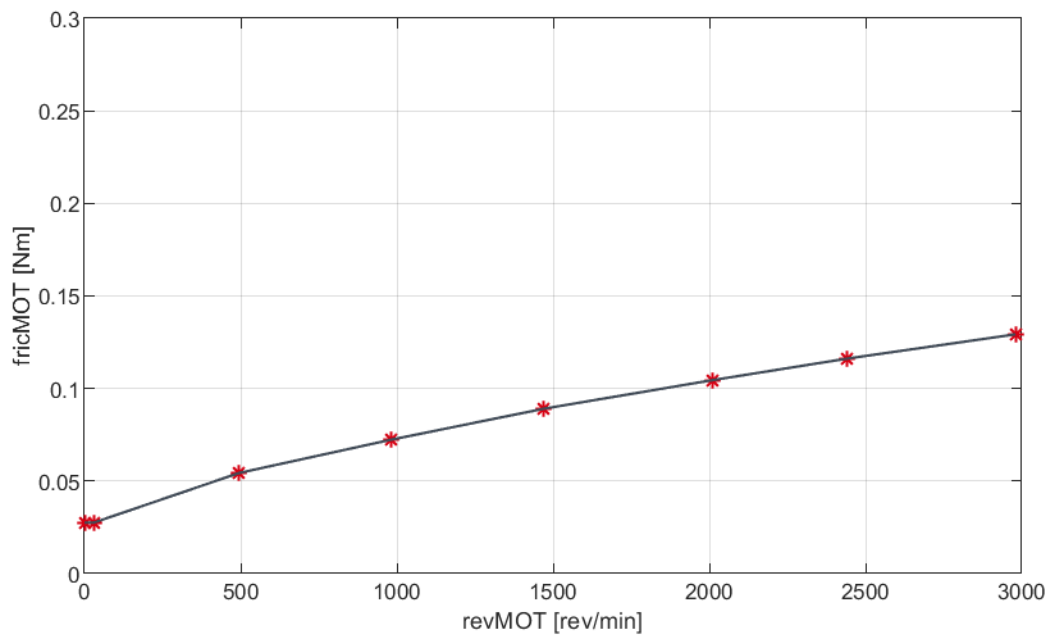




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4.1.1.2 Motor friction/damping characteristic

Information	
Motor	M82S
Manufacturer	Nidec
Parameter File	\\BOSCH.COM\DfsRB\DIV\VM\Engineering\EPS\CAE\ICN\task_data\EZB00470\03_Data\PMSM\014_NIDEC\M82S\2024_02_12\IDENTIFIKATION\ZFLSMOTOR50DEV.par





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4.2 Tuning Check TC

4.2.1 SWP/1/: Marginal Tune

Maneuver Catalogue:



Microsoft
Excel-Arbeitsblatt

SWP/1/: Marginal tune files from VT001

Vehicle Speed [km/h]	0	<10	>10	0
	Asphalt	Asphalt	Asphalt	Hoist
Center	 056_Marginal_Tuning.dcm	 060_Marginal_Tuning.dcm	 062_Marginal_Tuning.dcm	 066_Marginal_Tuning.dcm
Circle	 057_Marginal_Tuning.dcm	 064_Marginal_Tuning.dcm	 065_Marginal_Tuning.dcm	 069_Marginal_Tuning.dcm

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4.2.2 SWP/2/: Plant ID

Test data of VT002 is stored in the project folder of the CAE task. In addition to standard test catalogue, the test matrix was extended to include circle drive and hands on condition.

Plant Identification - open loop													
Test ID	Maneuver	Testing	General Conclions	Outside Temperature	ECU Temperature (ground plate)	v [kph]	PRBS Amplitude	Sine Amplitude*	Sine Frequency**	Bci min Gradient	Measurement number	DCM number	Comment
	Stand Still - on-center	Stand still measurement. Plant identification hands-off with steering wheel in center position	all limitation functions (DAC, Chatter etc) must be deactivated.	15°C	35°C	0		0.25Nm**	5Hz-45Hz	0.84	006_PlanID_SINE_01	006	
								0.4Nm	11Hz-43Hz		006_PlanID_SINE_02		
								0.5Nm	11Hz-43Hz		006_PlanID_SINE_03		
								0.6Nm	11Hz-43Hz		006_PlanID_SINE_04		
								1.5Nm	14Hz-100Hz		006_PlanID_SINE_05		
	Stand Still - off-center	Stand still measurement. Plant identification hands-off with steering wheel in off-center position (End-stop less 50deg)	all limitation functions (DAC, Chatter etc) must be deactivated.	15°C	35°C	0		0.25Nm**	5Hz-45Hz	0.785	007_PlanID_SINE_01	007	
								0.4Nm	11Hz-43Hz		007_PlanID_SINE_02		
								0.5Nm	11Hz-43Hz		007_PlanID_SINE_03		
								0.6Nm	11Hz-43Hz		007_PlanID_SINE_04		
								1.5Nm	14Hz-100Hz		007_PlanID_SINE_05		
	straight ahead driving - low vehSp	Driving straight ahead with vehicle speed below the limit which switches the FOC parameter (normally 10kph) Plant identification in hands-off condition	all limitation functions (DAC, Chatter etc) must be deactivated. Vehicle speed must be constant	15°C	40°C	8		0.25Nm**	5Hz-45Hz	0.785	008_PlanID_SINE_01	008	angle controller active
								0.4Nm	11Hz-43Hz		008_PlanID_SINE_02		
								0.5Nm	11Hz-43Hz		008_PlanID_SINE_03		
								0.6Nm	11Hz-43Hz		008_PlanID_SINE_04		
								1.5Nm	14Hz-100Hz		008_PlanID_SINE_05		
	straight ahead driving - high vehSp	Driving straight ahead with vehicle speed above the limit which switches the FOC parameter (normally 10kph) Plant identification in hands-off condition	all limitation functions (DAC, Chatter etc) must be deactivated. Vehicle speed must be constant	15°C	40°C	20kph>v>10kph (2nd gear idel) = 12 kph		0.25Nm**	5Hz-45Hz	1.005	009_PlanID_SINE_01	009	angle controller active, had to
								0.4Nm	11Hz-43Hz		009_PlanID_SINE_02		
								0.5Nm	11Hz-43Hz		009_PlanID_SINE_03		
								0.6Nm	11Hz-43Hz		009_PlanID_SINE_04		
								1.5Nm	14Hz-100Hz		009_PlanID_SINE_05		
	circle driving - low vehSp	Driving circle with vehicle speed below the limit which switches the FOC parameter (normally 10kph) Plant identification in hands-off condition	all limitation functions (DAC, Chatter etc) must be deactivated. Vehicle speed must be constant	15°C	40°C	7 kph, >240° SWA, 1.5 Nm angle controller		0.25Nm**	5Hz-45Hz	1.245	010_PlanID_SINE_01	010	angle controller active, split me
								0.4Nm	11Hz-43Hz		010_PlanID_SINE_02		
								0.5Nm	11Hz-43Hz		010_PlanID_SINE_03		
								0.6Nm	11Hz-43Hz		010_PlanID_SINE_04		
								1.5Nm	14Hz-100Hz		010_PlanID_SINE_05		
	circle driving - high vehSp	Driving circle with vehicle speed above the limit which switches the FOC parameter (normally 10kph) Plant identification in hands-off condition	all limitation functions (DAC, Chatter etc) must be deactivated. Vehicle speed must be constant	15°C	40°C	20kph>v>10kph (2nd gear idel) = 12 kph, >240° SWA, with angle controller (max 1.5 Nm)		0.25Nm**	5Hz-45Hz	1.7	011_PlanID_SINE_01	011	angle controller active, results c
								0.4Nm	11Hz-43Hz		011_PlanID_SINE_02		
								0.5Nm	11Hz-43Hz		011_PlanID_SINE_03		
								0.6Nm	11Hz-43Hz		011_PlanID_SINE_04		
								1.5Nm	14Hz-100Hz		011_PlanID_SINE_05		
	Hoist - on-center	Hoist measurement. Plant identification hands-off with steering wheel in center position	all limitation functions (DAC, Chatter etc) must be deactivated.	15°C	45°C	0		0.25Nm**	5Hz-45Hz	0.895	012_PlanID_SINE_01	012	angle controller active
								0.4Nm	11Hz-43Hz		012_PlanID_SINE_02		
								0.5Nm	11Hz-43Hz		012_PlanID_SINE_03		
								0.6Nm	11Hz-43Hz		012_PlanID_SINE_04		
								1.5Nm	14Hz-100Hz		012_PlanID_SINE_05		
	Hoist - off-center	Hoist measurement. Plant identification hands-off with steering wheel in off-center position (End-stop less 50deg)	all limitation functions (DAC, Chatter etc) must be deactivated.	15°C	45°C	0		0.25Nm**	5Hz-45Hz	0.825	013_PlanID_SINE_01	013	angle controller active
								0.4Nm	11Hz-43Hz		013_PlanID_SINE_02		
								0.5Nm	11Hz-43Hz		013_PlanID_SINE_03		
								0.6Nm	11Hz-43Hz		013_PlanID_SINE_04		
								1.5Nm	14Hz-100Hz		013_PlanID_SINE_05		

*piecewise constant frequency, frequ steps max 1Hz better 0.5Hz, 12 periods/frequency

**0.2Nm has to be adapted if mech.stop of TBT is reached (decrease value) or friction is too high (increase value)

***starting frequency adapted in case mech. End stop of TBT reached (decrease value)

Plant Identification - open loop - handsOn													
Test ID	Maneuver	Testing	General Concions	Outside Temperature	ECU Temperature	v [kph]	PRBS Amplitude	Sine Amplitude*	Sine Frequency**	Bci min Gradient	Measurement number	DCM number	Comment
0382-V702	Stand Still - on-center - handsOn (additional)	Stand still measurement. Plant identification hands-off with steering wheel in off-center position (End-stop less 50deg)	all limitation functions (DAC, Chatter etc) must be deactivated.	15°C	30°C	0		0.25Nm**	5Hz-45Hz	0.84	073 PlantID_SINE_01	007	localization partly moves the ha
								0.25Nm**	5Hz-45Hz		073 PlantID_SINE_01 (new test)		
								0.25Nm**	5Hz-45Hz		073 PlantID_SINE_01 (new test)		
								0.25Nm**	5Hz-45Hz		073 PlantID_SINE_02		
	Stand Still - off-center - handsOn (additional)	Stand still measurement. Plant identification hands-off with steering wheel in off-center position (End-stop less 50deg)	all limitation functions (DAC, Chatter etc) must be deactivated.	15°C	35°C	0		0.25Nm**	5Hz-45Hz	0.84	073 PlantID_SINE_03	007	
								0.25Nm**	5Hz-45Hz		073 PlantID_SINE_03 (new test)		
								0.25Nm**	5Hz-45Hz		073 PlantID_SINE_03 (new test)		
								0.25Nm**	5Hz-45Hz		073 PlantID_SINE_04		
	straight ahead driving - low vehSp - handsOn (additional)	Driving straight ahead with vehicle speed below the limit which switches the FOC parameter (normally 10kph) Plant identification in hands-off condition	all limitation functions (DAC, Chatter etc) must be deactivated. Vehicle speed must be constant	15°C	40°C	8		0.25Nm**	5Hz-45Hz	0.84	073 PlantID_SINE_05	007	no angle controller active
								0.25Nm**	5Hz-45Hz		073 PlantID_SINE_05 (new test)		
								0.25Nm**	5Hz-45Hz		073 PlantID_SINE_05 (new test)		
								0.25Nm**	5Hz-45Hz		073 PlantID_SINE_06		
straight ahead driving - high vehSp - handsOn (additional)	Driving straight ahead with vehicle speed above the limit which switches the FOC parameter (normally 10kph) Plant identification in hands-off condition	all limitation functions (DAC, Chatter etc) must be deactivated. Vehicle speed must be constant	15°C	40°C	20kph<v<10kph (2nd gear idle)		0.25Nm**	5Hz-45Hz	1.005	073 PlantID_SINE_07	007	no angle controller active	
							0.25Nm**	5Hz-45Hz		073 PlantID_SINE_07 (new test)			
							0.25Nm**	5Hz-45Hz		073 PlantID_SINE_07 (new test)			
							0.25Nm**	5Hz-45Hz		073 PlantID_SINE_08			
circle driving - low vehSp - handsOn (additional)	Driving circle with vehicle speed below the limit which switches the FOC parameter (normally 10kph) Plant identification in hands-off condition	all limitation functions (DAC, Chatter etc) must be deactivated. Vehicle speed must be constant	15°C	30°C	7 kph, <220° SWA, with angle controller (max 1.5 Nm)		0.25Nm**	5Hz-45Hz	1.245	073 PlantID_SINE_09	007	with angle controller for margin	
							0.25Nm**	5Hz-45Hz		073 PlantID_SINE_09 (new test)			
							0.25Nm**	5Hz-45Hz		073 PlantID_SINE_09 (new test)			
							0.25Nm**	5Hz-45Hz		073 PlantID_SINE_10			
circle driving - high vehSp - handsOn (additional)	Driving circle with vehicle speed above the limit which switches the FOC parameter (normally 10kph) Plant identification in hands-off condition	all limitation functions (DAC, Chatter etc) must be deactivated. Vehicle speed must be constant	15°C	35°C	20kph<v<10kph (2nd gear idle) = 12 kph, <220° SWA, with angle controller (max 1.5 Nm)		0.25Nm**	5Hz-45Hz	1.7	073 PlantID_SINE_11	007	with angle controller for margin	
							0.25Nm**	5Hz-45Hz		073 PlantID_SINE_11 (new test)			
							0.25Nm**	5Hz-45Hz		073 PlantID_SINE_11 (new test)			
							0.25Nm**	5Hz-45Hz		073 PlantID_SINE_12			
Hoist - on-center - handsOn (additional)	Hoist measurement. Plant identification hands-off with steering wheel in center position	all limitation functions (DAC, Chatter etc) must be deactivated.	15°C	45°C			0.25Nm**	5Hz-45Hz	0.895	081 PlantID_SINE_01	008	with angle controller	
							0.25Nm**	5Hz-45Hz		081 PlantID_SINE_01 (new test)			
							0.25Nm**	5Hz-45Hz		081 PlantID_SINE_01 (new test)			
							0.25Nm**	5Hz-45Hz		081 PlantID_SINE_02			
Hoist - off-center - handsOn (additional)	Hoist measurement. Plant identification hands-off with steering wheel in off-center position (End-stop less 50deg)	all limitation functions (DAC, Chatter etc) must be deactivated.	15°C	45°C	0 kph, <210° SWA left with angle controller		0.25Nm**	5Hz-45Hz	0.825	082 PlantID_SINE_03	008	with angle controller	
							0.25Nm**	5Hz-45Hz		082 PlantID_SINE_03 (new test)			
							0.25Nm**	5Hz-45Hz		082 PlantID_SINE_03 (new test)			
							0.25Nm**	5Hz-45Hz		082 PlantID_SINE_04			

*piecewise constant frequency, frequ steps max 1Hz better 0.5Hz, 12 periods/frequency

handsOn: use same sign as handsOff

HandsOn: put yellow on door, place the hand flat on the steering wheel



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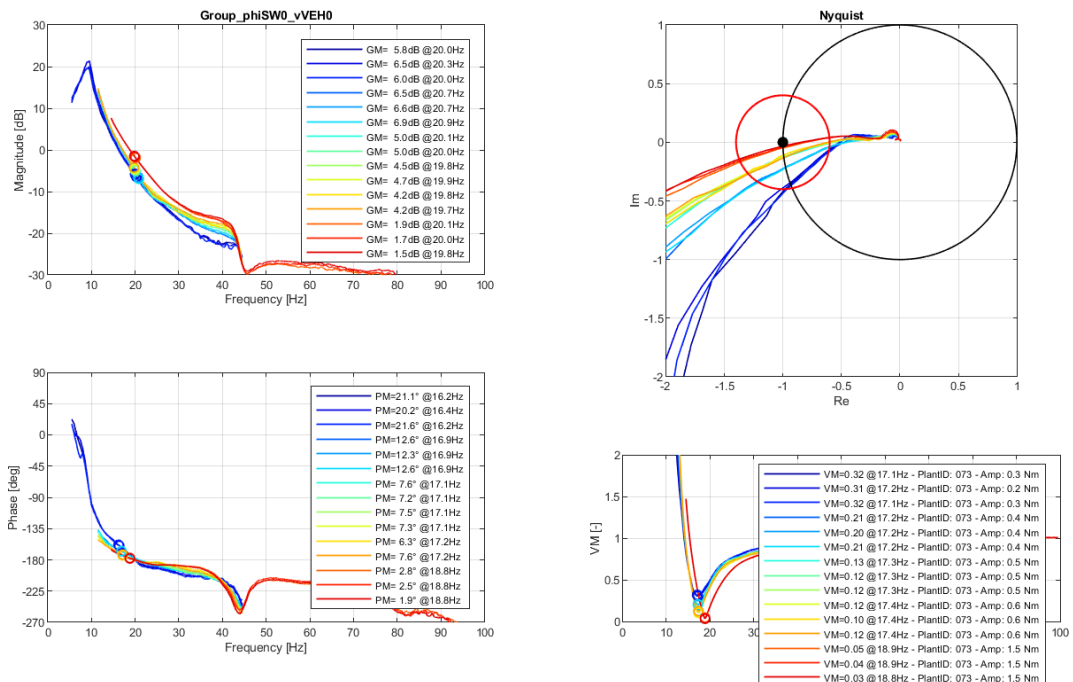
4.2.3 SWP/3/: Identification of plant and marginal tune validation

	Asphalt	Hoist
Hands On	CAE2600637.pptx	CAE2600635.pptx
Hands Off	CAE2600638.pptx	CAE2600636.pptx

➔ Test data is valid and marginal tune validation is successful (VM values are small)

Remarkable repeatability.

On asphalt hands on 0km/ several tests were repeated and show remarkable low variation.





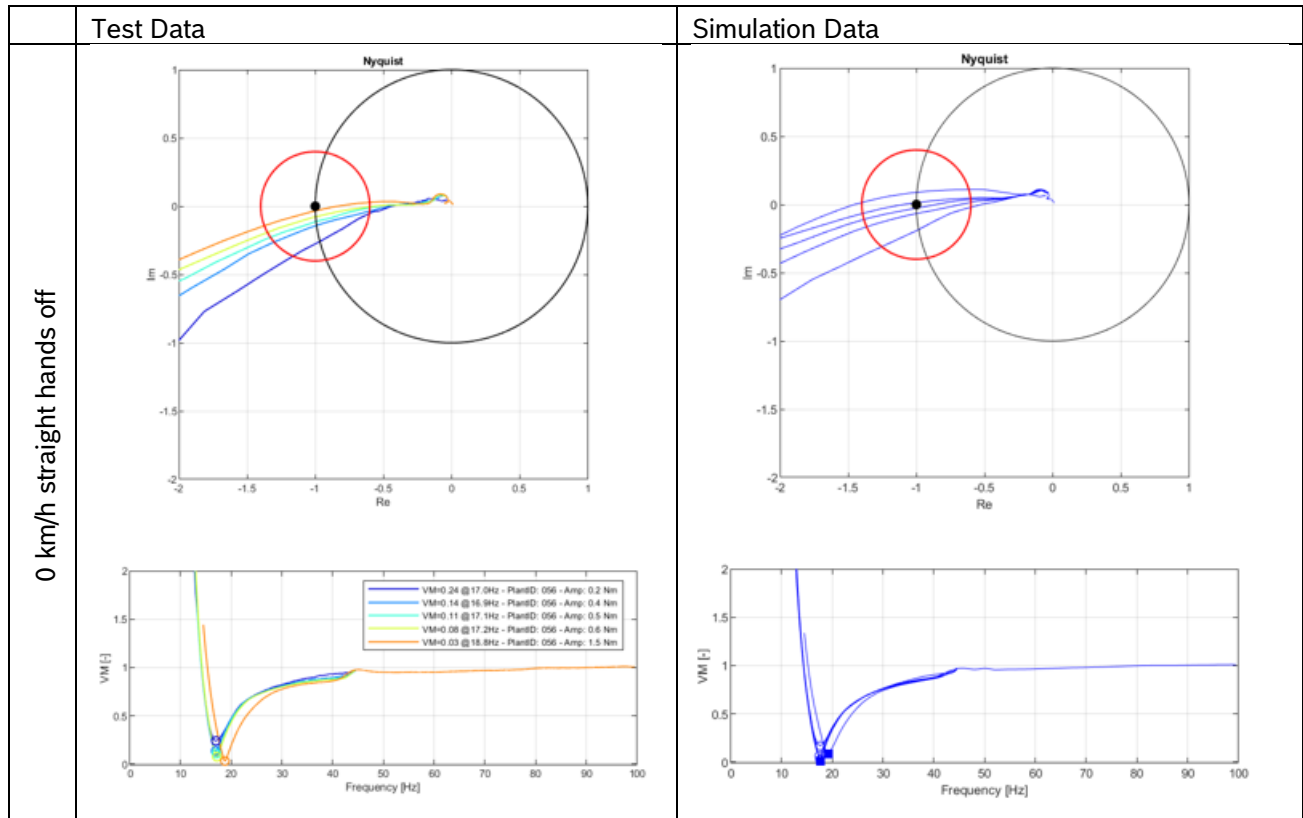
Calculation Report 0040499642_001 [00]

4.2.4 SWP/4/: Controls ID and Validation

Marginal tune validation reveals inaccuracy due to AUTOSAR parameter mismatch between vECU and vehicle test. Simulation results tend to be slightly more critical. One Example is shown below. Validation results for different marginal tunes and boundary conditions are shown in the appendix [1].

Examples Asphalt Hands Off 0km/h – Straight – $K_{p_crit} = 0,84$:

Obviously, simulation VM values are more conservative than test data. Critical Frequencies in 18-20Hz match.



Decision is to continue with SWP/5/ because tuning data in SW 450_700_C_08_00 includes all customer specific application data and is included in vECU without modification of AUTOSAR values.



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4.2.5 SWP/5/: Stability DOE

SWP/5/ was completed, even though SWP/4/ failed due to AUTOSAR parameter mismatch between test and simulation. Customer tuning is expected to be the same for standard tuning 450_700_C_08_00.

No unstable regions are found in the VM maps.

Asphalt most Critical Condition: Hands off – Center - Slow driving <10 km/h - TBT > 1.2Nm

Hoist most Critical Condition: Hands off – Center - Slow driving 10-30 km/h - TBT > 1.2Nm

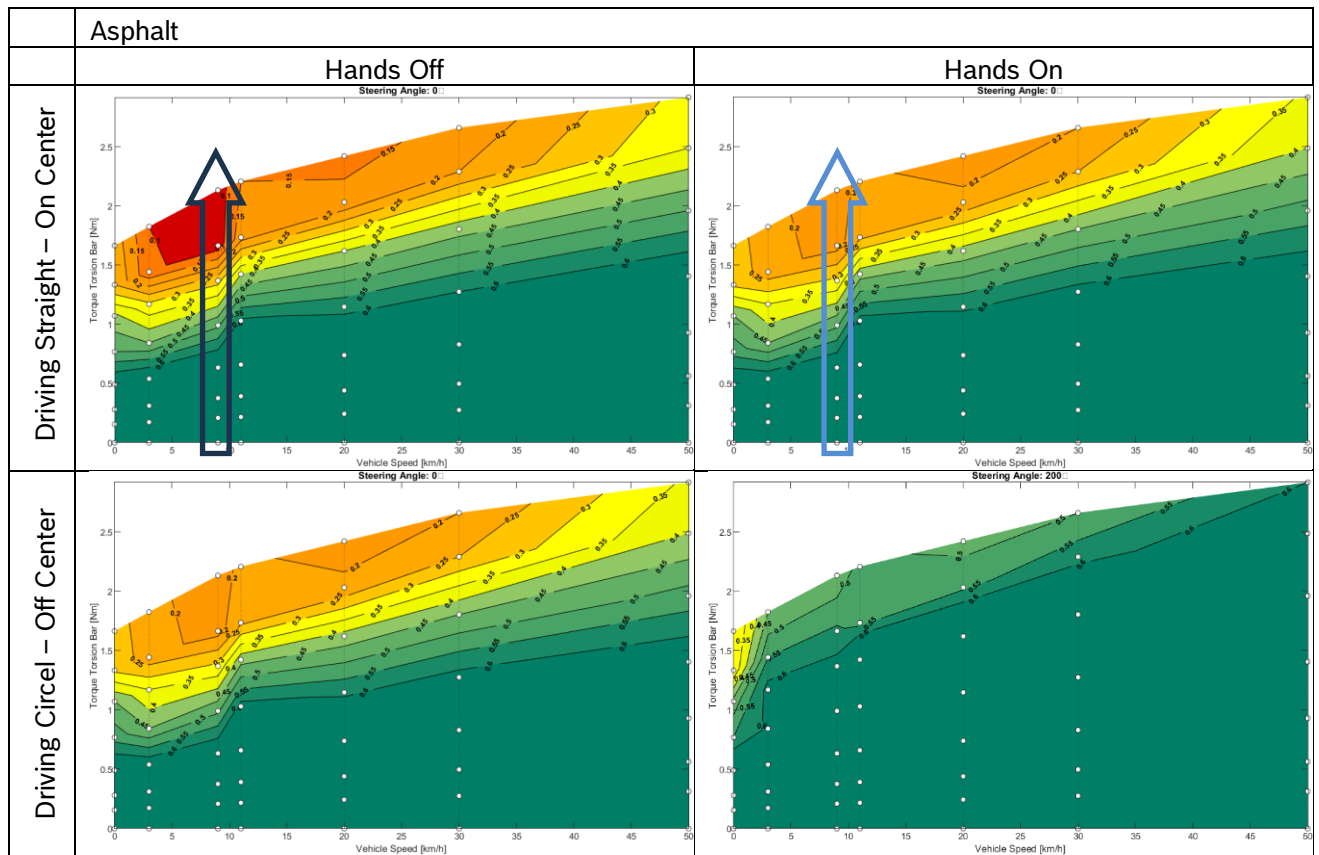
Summary:

- Hands off more critical than hands on
- On center more critical than off center
- Hoist more critical than asphalt

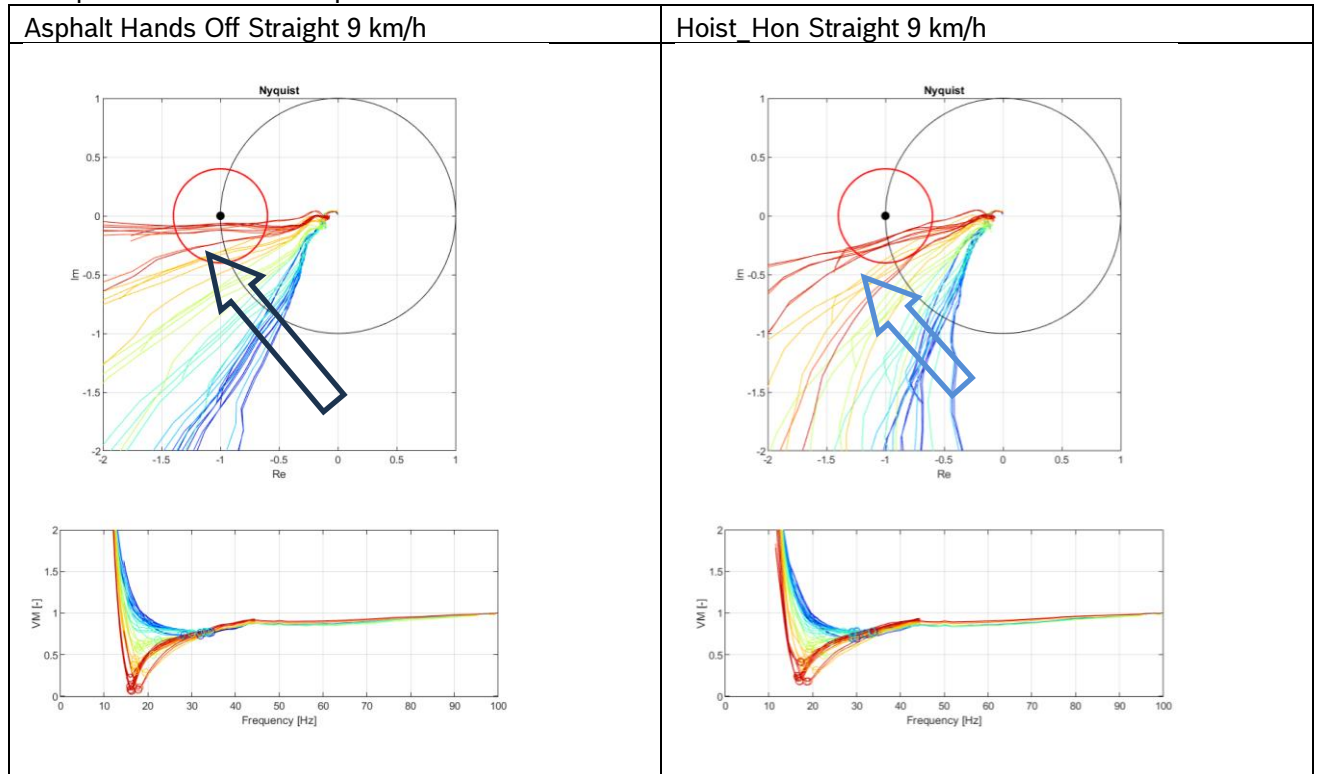
Lowest VM values are in the frequency range 17-20Hz



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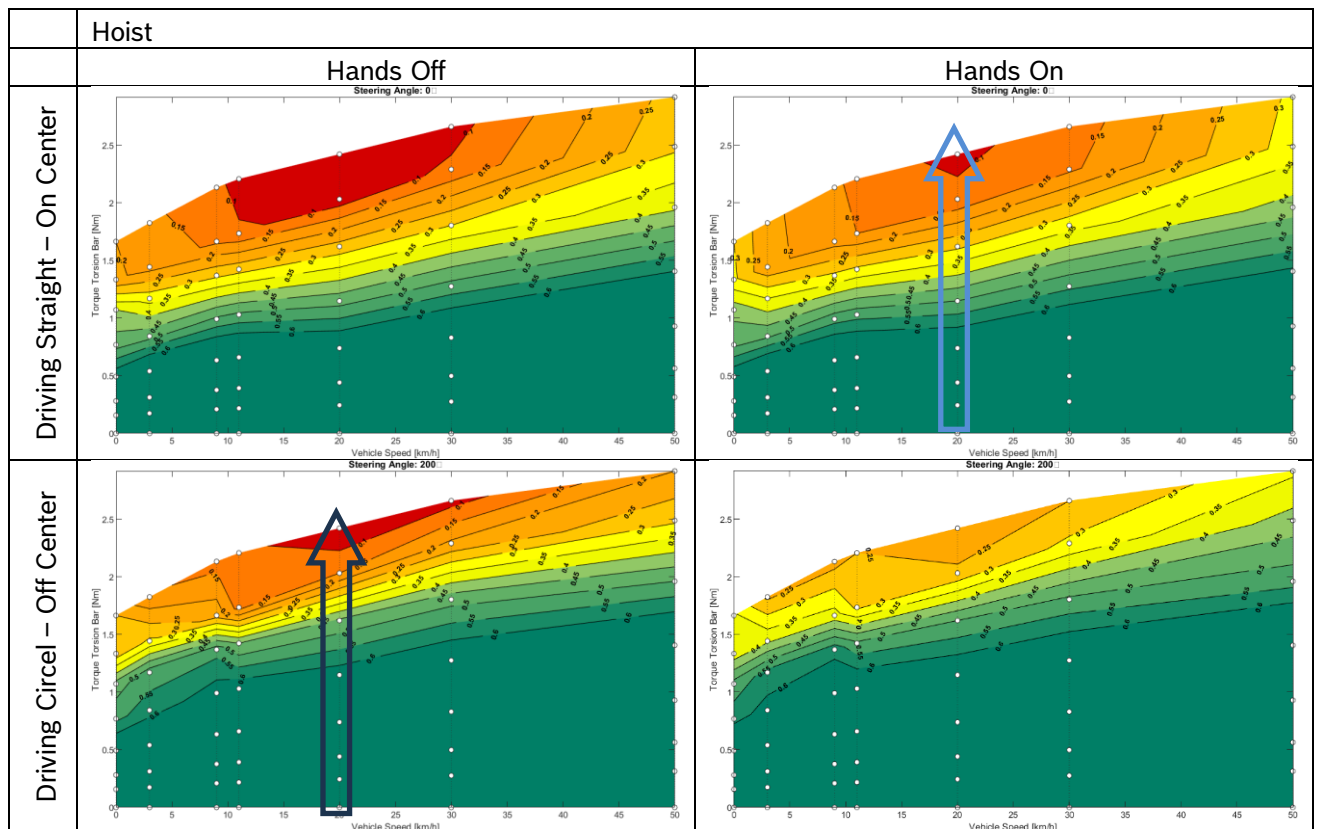
Examples from the heat maps



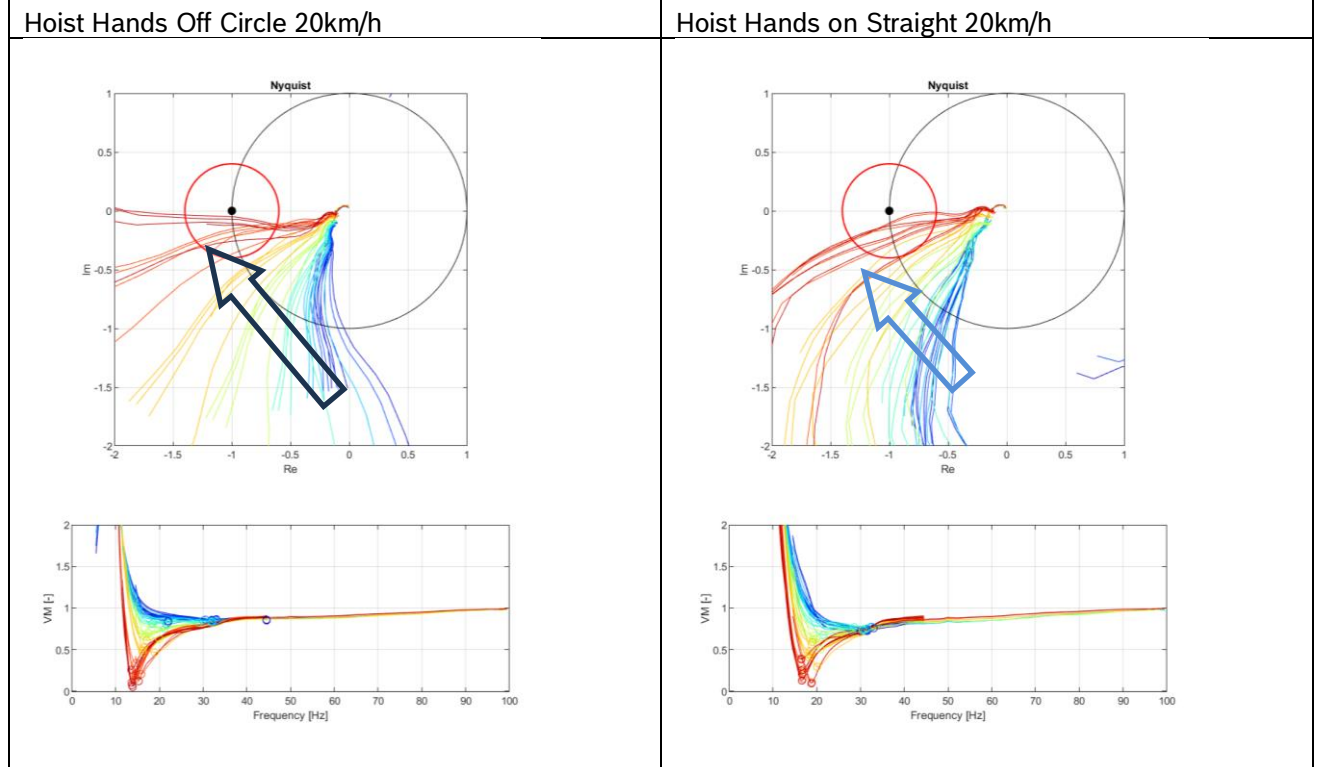
● Fulfills requirement /
 ● Unclear if requirement fulfilled /
 ● Violates requirement



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Examples from the heat maps



4.2.6 SWP/6/: Tuning validation

The final on-vehicle validation for the most critical conditions identified in simulation was not performed.

Fulfils requirement / Unclear if requirement fulfilled / Violates requirement



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4.3 Appendix

More comprehensive representation of all results: [CAE2600623.pptx](#)